Debugging and Monitoring LLMs in Production

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About Abi Aryan

Abi Aryan is the founder of Abide AI and a machine learning engineer with over eight years of experience in the ML industry building and deploying machine learning models in production for recommender systems, computer vision, and natural language processing—within a wide range of industries such as e-commerce, insurance, and media and entertainment.

Previously, she was a visiting research scholar at the Cognitive Sciences Lab at UCLA where she worked on developing intelligent agents. She has also authored research papers in AutoML, multi-agent systems, and LLM cost modeling and evaluations.

Books:

- LLMOps: Managing Large Language Models in Production, O'Reilly Publications (July 2025)
- GPU Engineering for AI Systems, Packt Publications (Sept 2026)

Agenda

01. Why - Understanding the Challenges
02. What - Monitoring & Debugging Techniques
03. How - Tooling & Feedback Loops

01. Understanding the Challenges

Evaluating LLMs is hard

1. **Scale of the problem**: LLMs are trained on massive amounts of data, and the number of possible inputs they can receive is practically infinite.

   This makes it impossible to exhaustively test them on every scenario, unlike simpler programs. Even evaluating a tiny fraction of possibilities is a monumental task.

Some of the possibilities would be:-

- ❏ **Informativeness & Factuality** - *Assessing how well the outputs are informative and correspond to factual information.*
- ❏ **Fluency & Coherence** - *Measuring how well the outputs are grammatically correct, readable, and follow a logical flow.*
- ❏ **Engagement & Style** - *Evaluating how engaging and interesting the LLM's outputs are, along with stylistic aspects.*
- ❏ **Safety & Bias** - *Potential biases or harmful content generated by the LLM.*
- ❏ **Grounding** - *Assessing how well the LLM's response is grounded in real-world information and avoids hallucinations.*
- ❏ **Efficiency** - *Measuring the computational resources required by the LLM to generate outputs.*

Evaluating LLMs is hard

2. Defining "good": Unlike some models where there's a clear success metric (think image recognition accuracy), what constitutes a "good" response from an LLM can be subjective. Is it providing relevant information? Is it creative? Is it factually accurate? These goals can conflict, making it hard to design a single metric that captures everything.

“Good performance” can mean several things-

- **Task-Specific Success:** *An LLM's "goodness" is highly dependent on the task it's designed for.*
- **Accuracy and Factuality:** *For tasks like question answering or summarizing topics, an LLM should be demonstrably accurate and avoid generating false or misleading information.*
- **Fluency and Coherence:** *The language should be appropriate for the context and audience.*
- **Relevance and Informativeness:** *The LLM's response should be relevant to the prompt or query and provide useful information. It shouldn't go off on tangents or introduce irrelevant details.*
- **Engagement and Creativity:** *Depending on the context, "good" might mean generating outputs that are interesting, engaging, or even surprising.*
- **Safety and Fairness:** *An LLM shouldn't generate harmful content, promote biases, or perpetuate stereotypes. It should be fair and inclusive in its responses.*
- **Efficiency:** *Ideally, an LLM should be able to generate good outputs while using computational resources efficiently. This becomes important for real-world applications where processing power is limited.*

LLM Evaluation Criteria

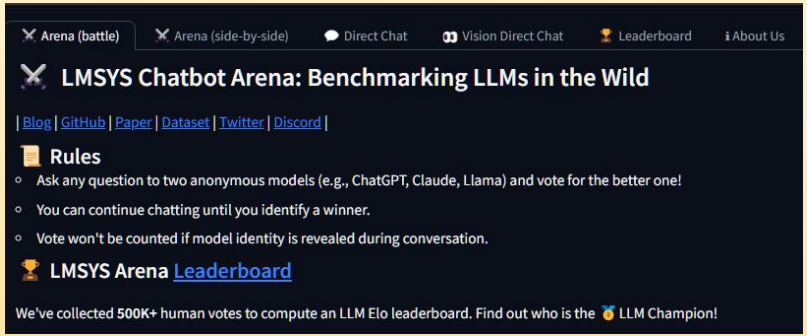
Evaluation Criteria	**Broad goals for LLM performance**	**Task Understanding, Reasoning Capabilities, Appropriateness, Safety**
Language Features	Focus on core language skills	Fluency, Coherence, Factuality
Task Independence	Metrics applied across various tasks	Toxicity, Fairness, Bias
Task Dependence	Metrics specific to a particular task	Relevance (Question Answering), Appropriateness (Machine Translation)
Metrics	Quantitative measures of performance	BLEU (Machine Translation), ROUGE (Summarization), Accuracy (Question Answering)
LLM Specific Metrics	Pre-defined datasets and metrics for specific tasks	GLUE, SuperGLUE, HellaSwag
Custom Metrics	User-defined metrics based on specific needs	Guideline Adherence, Presence of Certain Words
Frameworks	Tools and platforms for conducting evaluations	Ragas, Promptfoo, RAG Triad
Human Evaluation	Judgments by human experts (considered Gold Standard)	LMSys Arena

Some Limitations

- **Human Evaluation Limitations**: While Human Evaluation is considered the gold standard, it can be expensive, time-consuming, and prone to subjectivity.

- **Benchmark Limitations**: Pre-defined benchmarks can be susceptible to reverse-engineering, where the model learns to perform well on the benchmark without necessarily generalizing to real-world tasks.

- **LLM Evaluator Limitations:** Powerful but potentially biased, opaque, and resource-intensive.



BLEU (Machine Translation), ROUGE (Summarization), Accuracy (Question Answering)

GLUE, SuperGLUE, HellaSwag

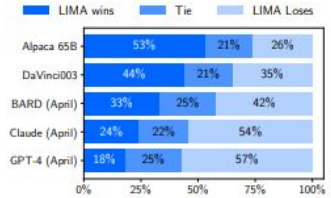


Figure 1: Human preference evaluation, comparing LIMA to 5 different baselines across 300 test prompts.

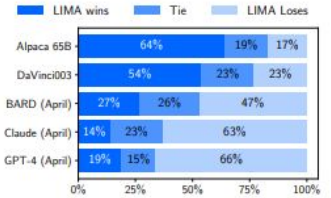
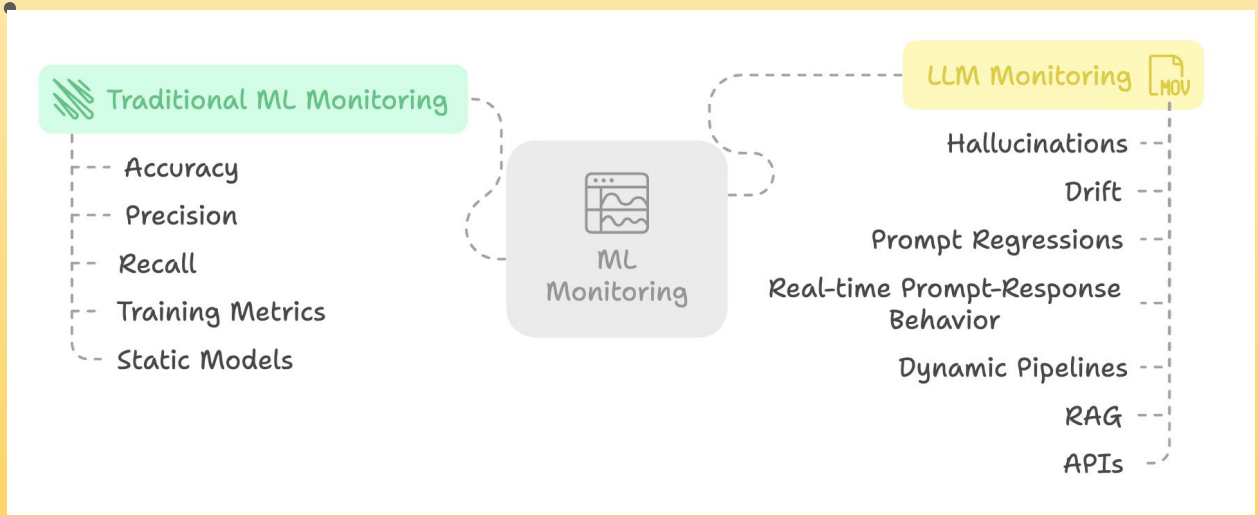


Figure 2: Preference evaluation using GPT-4 as the annotator, given the same instructions provided to humans.

Why “Observability” matters

LLMs are powerful but unpredictable. **Observability ensures control, safety, and performance in real-world use aka pipelines.**

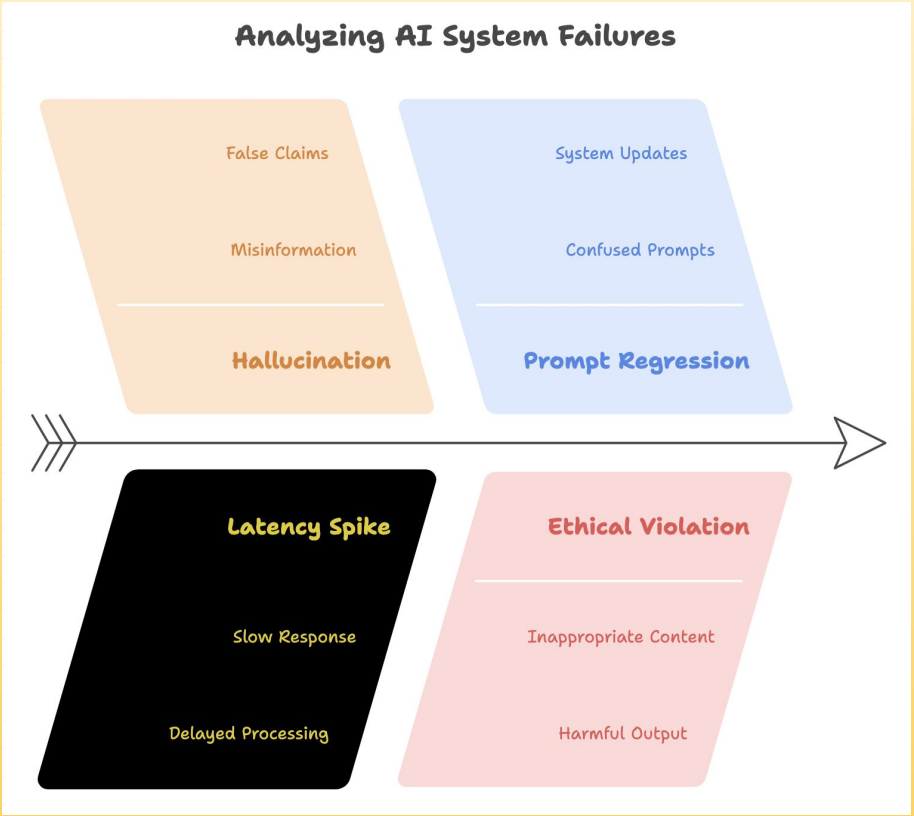
1. LLMs are non-deterministic
2. Failure is inevitable in production
3. LLMs interact with real users
4. Traditional ML monitoring ≠ enough
5. Regulatory and ethical scrutiny is rising

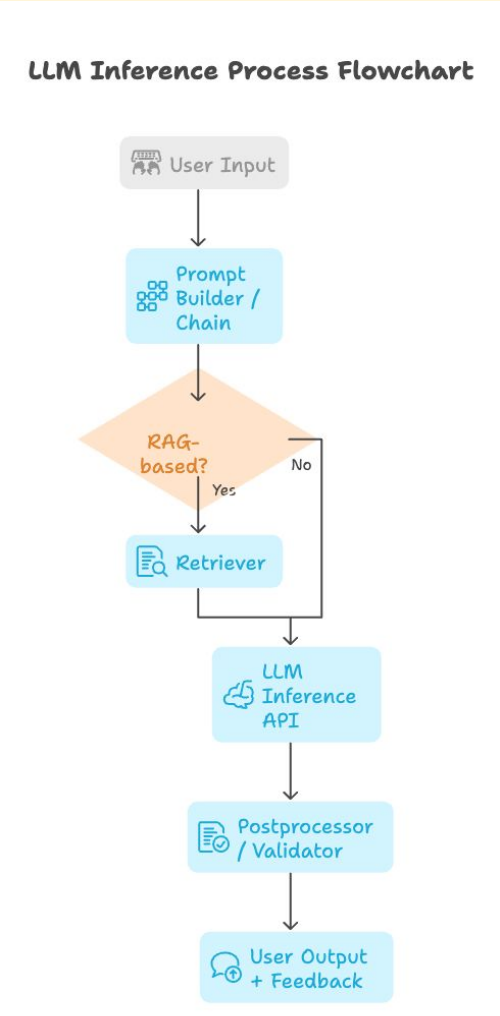


Common Failure Modes in LLM Pipelines

LLMs break in subtle and not-so-subtle ways and knowing **where and how** helps you detect issues early.

1. **Hallucinations**
2. **Prompt Regressions**
3. **Latency Spikes**
4. **Data Drift**
5. **Inconsistent Behavior**
6. **Ethical & Compliance Risks**





Key Components of an LLM Pipeline

Input Interface - Unexpected formats, malicious content

Preprocessing / Orchestration Layer -Prompt bugs, formatting errors, missing context

Retriever (Optional for RAG) - Retrieval latency, irrelevant or outdated context

LLM Inference -: Hallucinations, long responses, API instability

Post Processing - Structure mismatches, broken JSON, empty completions

User-Facing Output + Feedback Capture - Risk: User misunderstanding, missing feedback data

02. Monitoring & Debugging Techniques

What to Monitor – Observability Categories

User Input & Prompts

Monitor formatting and injections

Retrieval

Monitor relevance and latency

LLM Inference Layer

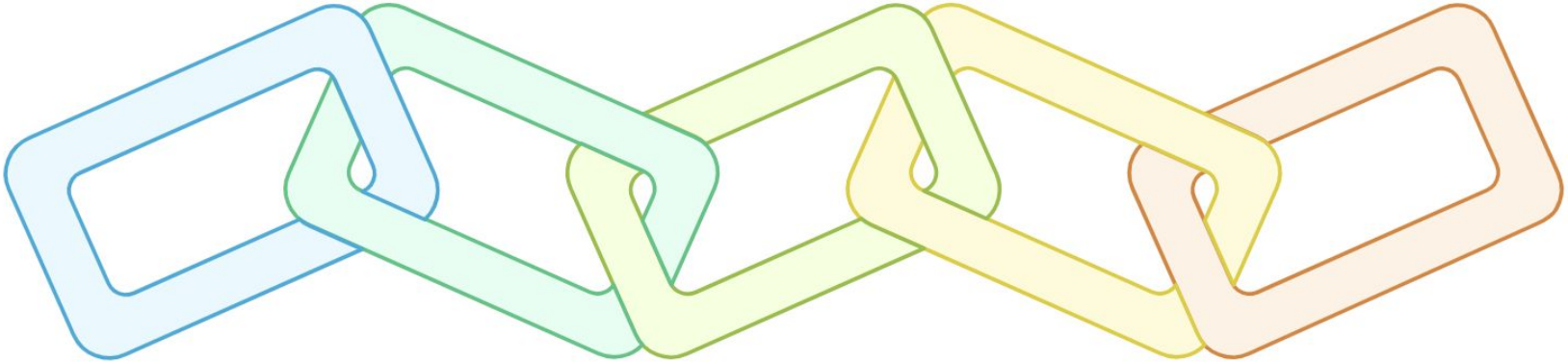
Monitor performance, cost, and errors

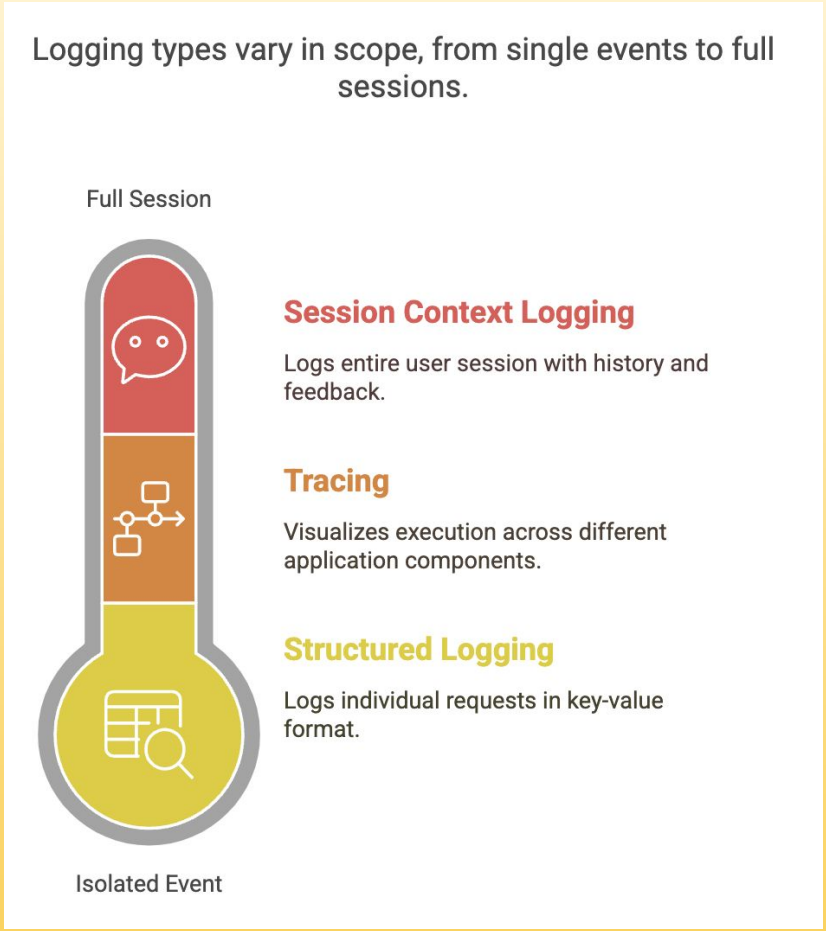
Output Parsing

Monitor JSON/schema conformance

User Interaction Layer

Monitor feedback and satisfaction



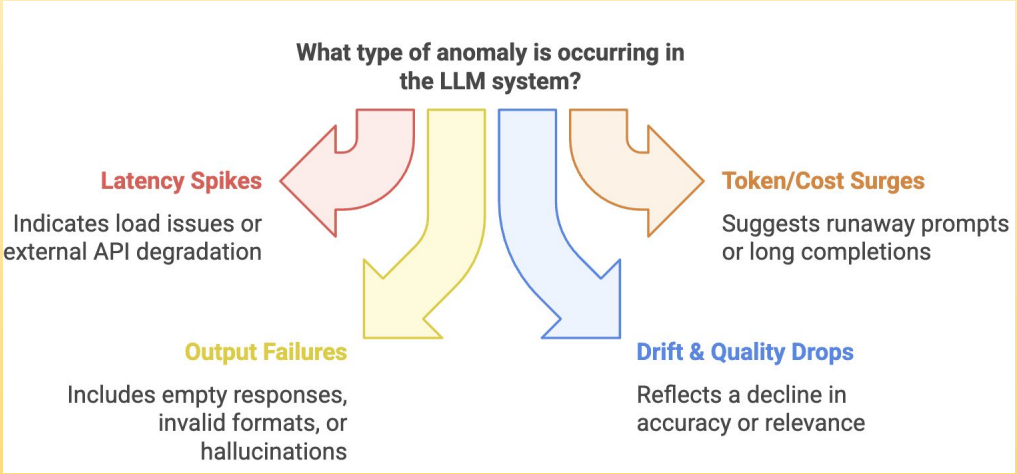


Structured Logging & Tracing for LLMs

Best Practices:

- Standardize log schemas across services
- Include version info (model, prompt, retriever, chain)
- Sanitize logs for PII before storing
- Use UUIDs to correlate inputs and outputs

Anomaly Detection in Production



Some methods:

1. **Threshold-Based Alerts**

e.g., response_time > 2s, token_count > 1024

2. **Statistical Anomaly Detection**

Rolling averages, standard deviations, z-scores

3. **Drift Detection**

    a. Monitor input distribution (e.g., query types)

    b. Monitor embedding similarity distributions (for retriever drift)

4. **Feedback Signal Analysis**

Sudden drop in thumbs-up ratio or user ratings

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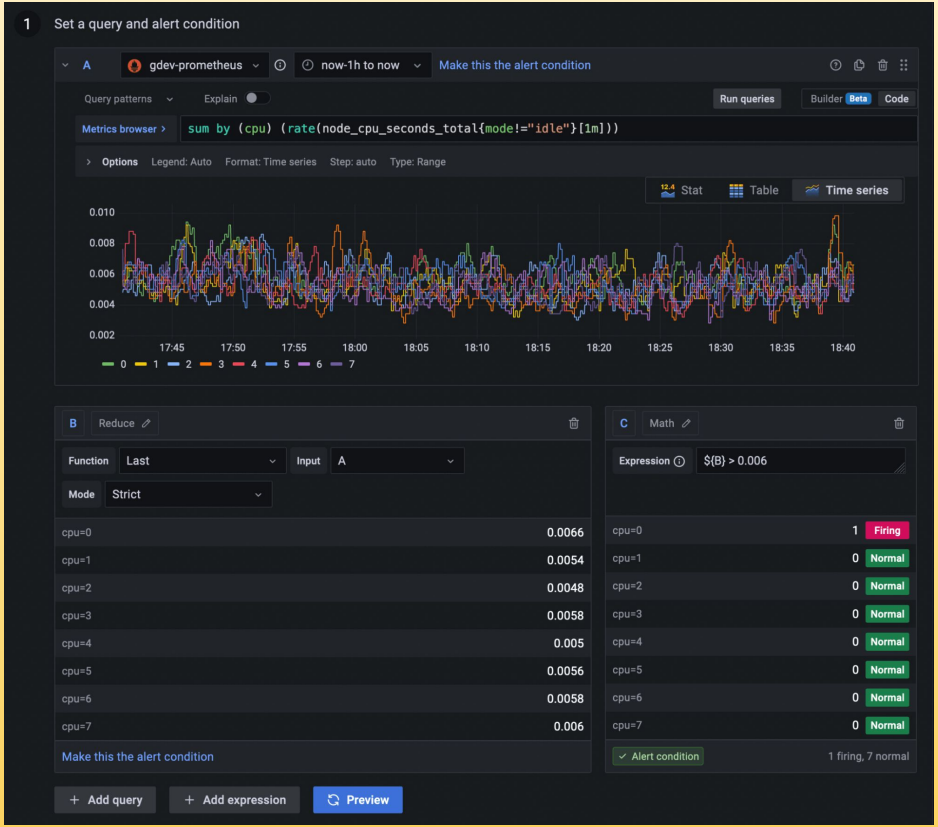

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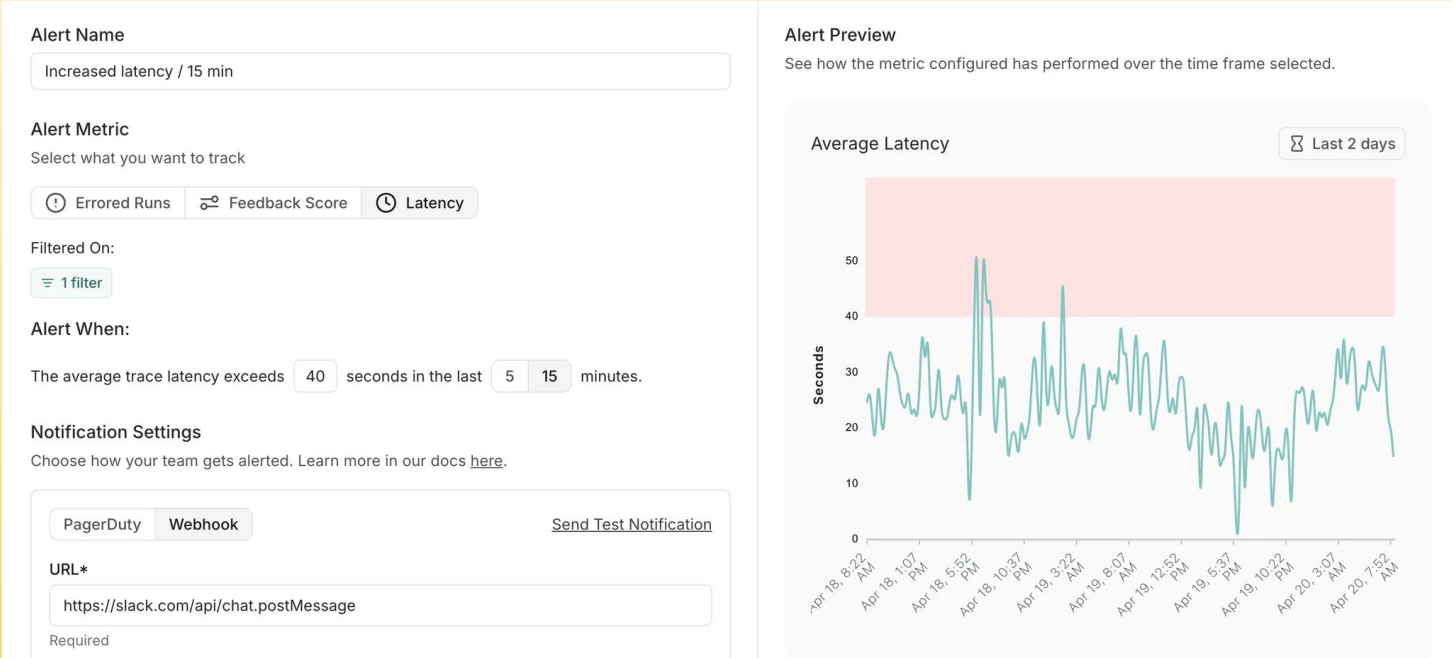

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Anomaly Detection in Production

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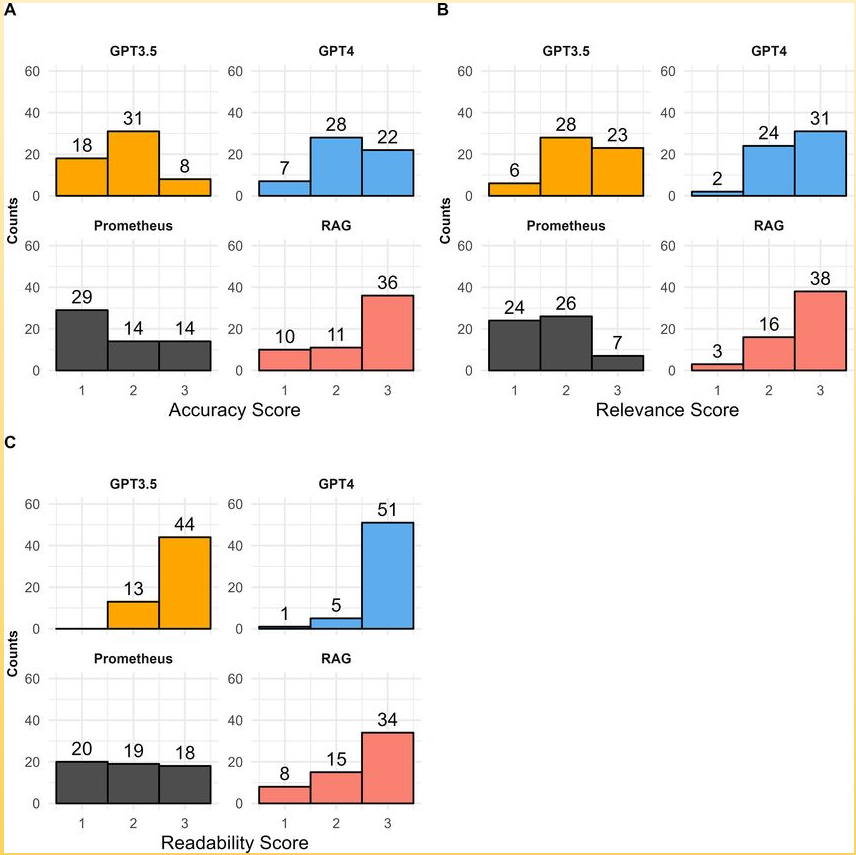

Anomaly Detection in Production

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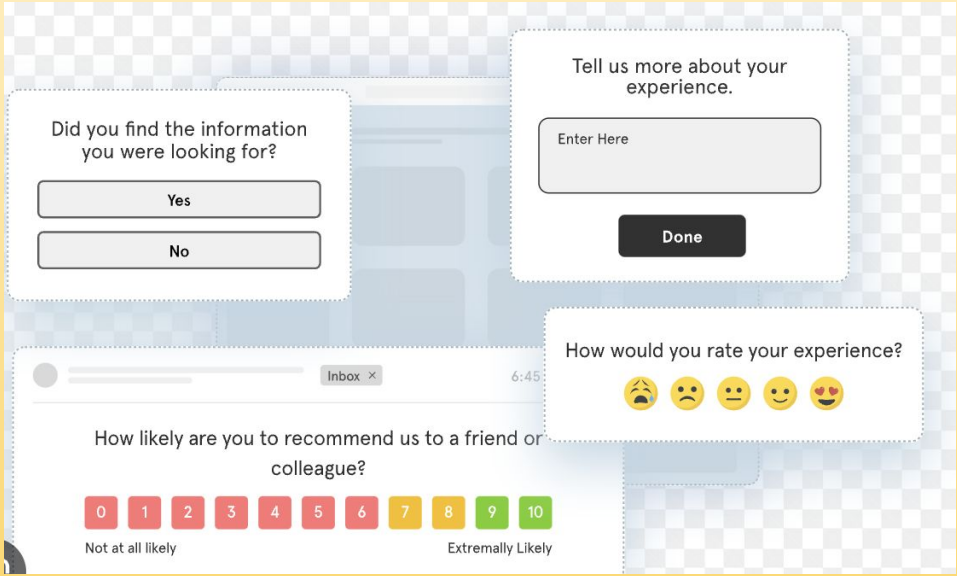


03. Tooling & Feedback Loops

Metrics That Matter – Performance, Drift, Hallucination, Ethics

Performance Metrics

Latency (avg, p95, p99)

Token usage (input/output split, cost tracking)

Throughput (requests per second)

Timeouts & retries

Completion length distribution

Data & Query Drift

Input drift: Changes in prompt shapes, user query types

Retriever drift: Shift in similarity scores, relevance of retrieved docs

Embedding drift: Embedding vector distribution shifts (esp. across model upgrades)

Hallucination & Output Quality

Factual accuracy (e.g., using eval sets or ground truth)

Consistency across reruns (same input → different answers?)

Response structure correctness (JSON, schema conformance)

Ethical & Safety Metrics

Toxicity levels

Bias detection (gender, race, nationality)

PII leakage detection

Safety classification scores

Different LLM architectures demand different observability strategies

RAG (Retrieval-Augmented Generation)

Monitor retrieval relevance

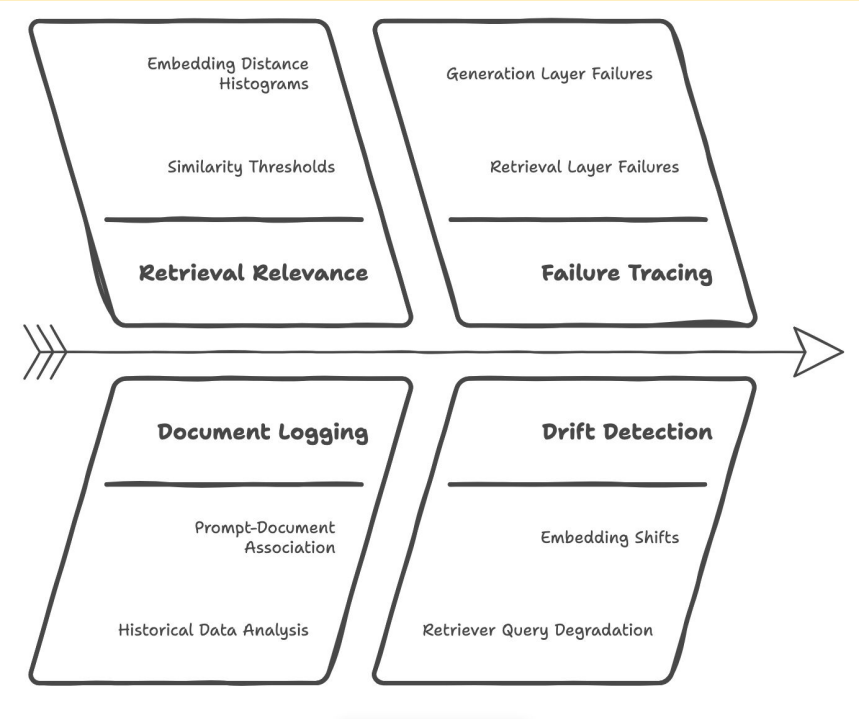
- Use similarity thresholds and embedding distance histograms
- Auto-evaluate retrieved docs against ground truth

Log retrieved documents with each prompt

Trace failures to retrieval OR generation layer

Drift Detection: Embedding shifts, retriever query degradation

LangSmith + Vector Store Logs + Open Telemetry



Different LLM architectures demand different observability strategies

Chatbots

Session-based tracing: Track user flow, turn-by-turn behavior

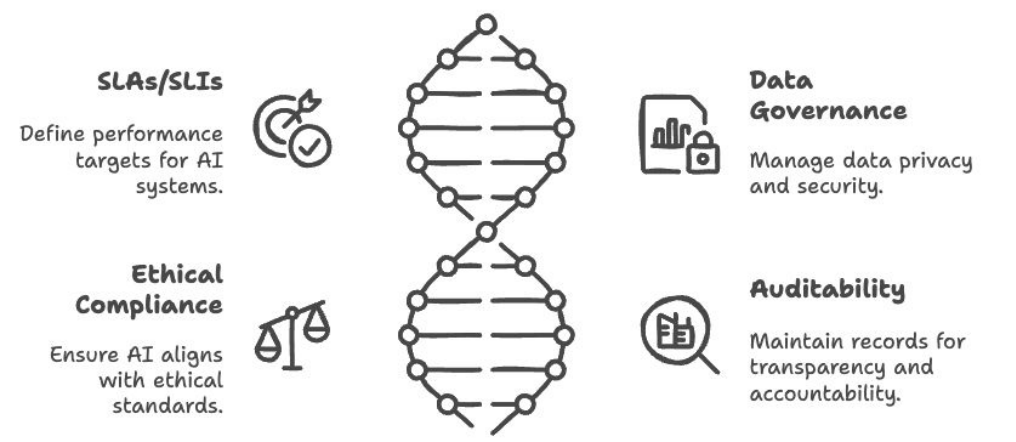
Escalation metrics: How often users retry or escalate to human agents

Toxicity & safety filters at response and prompt level

Feedback-driven improvement loop

LangSmith + Structured Logging + feedback dashboards

Different LLM architectures demand different observability strategies



Enterprise AI Applications

SLAs/SLIs: Define hard targets (e.g., response under 1s, 99.9% uptime)

Data governance logging: Log user input anonymization, PII redaction

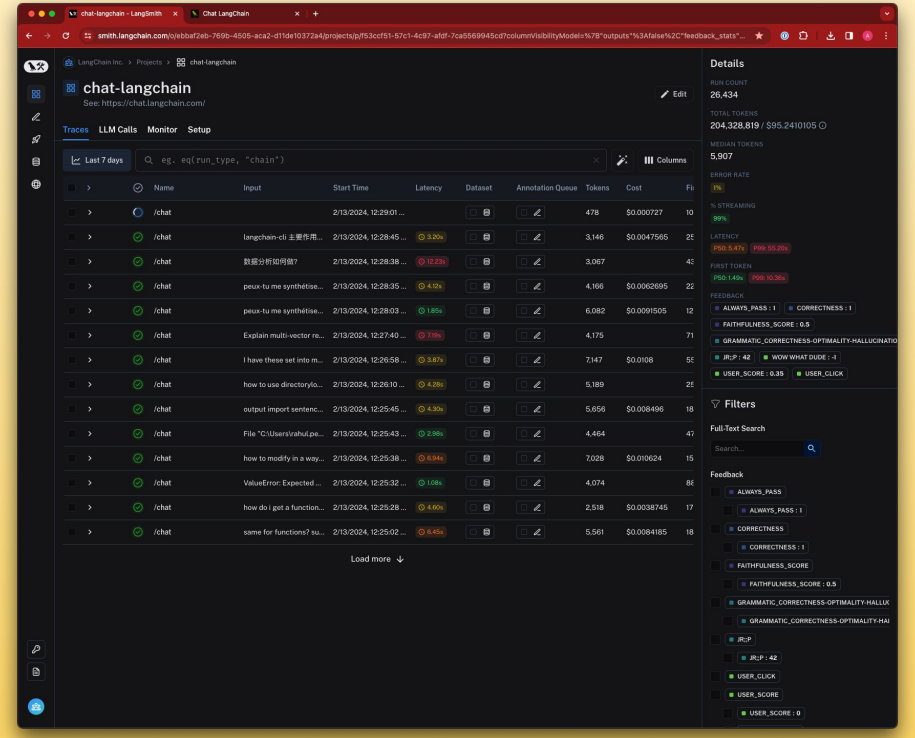
Ethical compliance checks: Model outputs scored against company risk frameworks

Auditability: Retain trace logs for inspection

Grafana/Prometheus + MLflow/ZenML + internal logging frameworks

Some “in-general” best practices

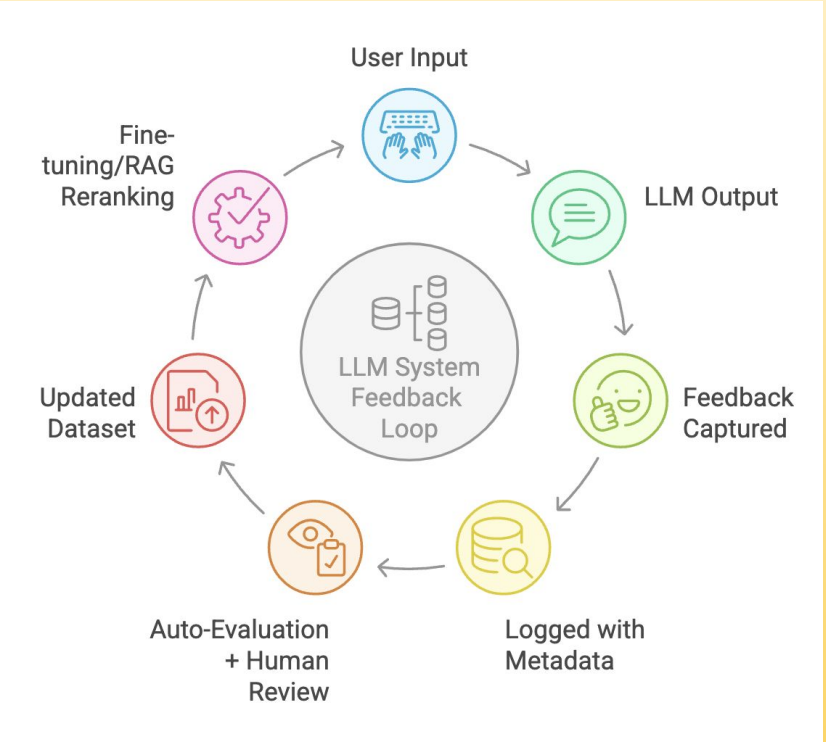
- Use **LangSmith** or similar to trace full LLM pipeline execution
- Store logs with metadata: model version, prompt version, retriever version
- Include **user feedback hooks** for all user-facing LLM systems
- Tag and version all experiments — models, prompts, context logic



And finally, Feedback loops

3 types

1. **Explicit Feedback**
   - User thumbs-up/down, star ratings, comments
   - Helps label data for fine-tuning or re-ranking
   - Use in LangSmith or in-house dashboards
2. **Implicit Feedback**
   - User dwell time, retries, reformulations, click-throughs
   - Track where users abandon or escalate the task
3. **System Feedback**
   - Eval scores (e.g., accuracy, structure validity)
   - Logs flagged for anomalies or hallucinations



Thank you! *Time for Q & A?*

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www.abiaryan.com